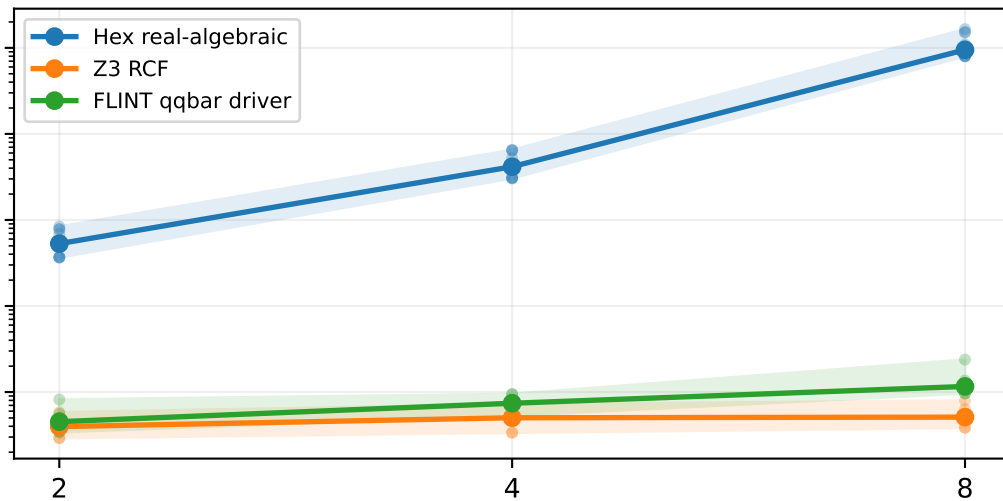


Actual real-algebraic polynomial roots: Hex, Z3 RCF and FLINT qqbar

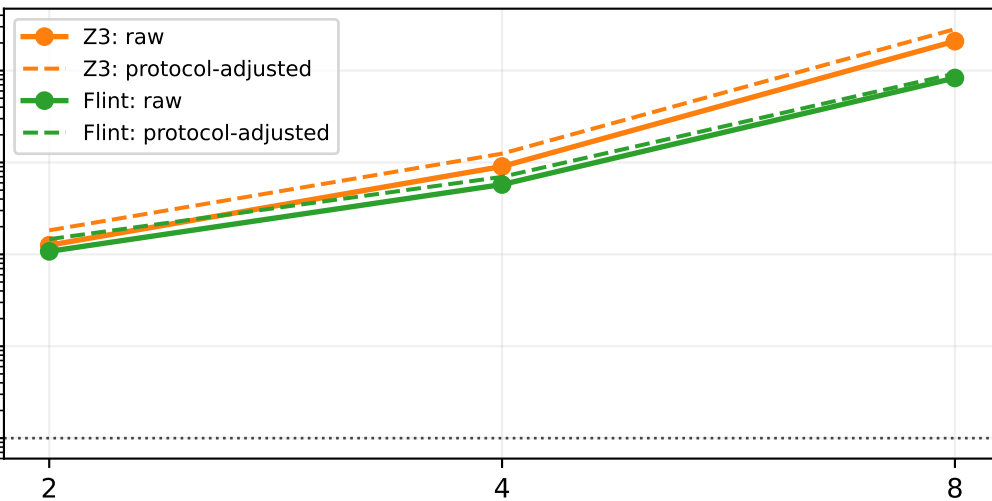
$$X^n - 2$$

Root API + result checks (ms, log scale)



Polynomial degree n (coefficient height fixed)

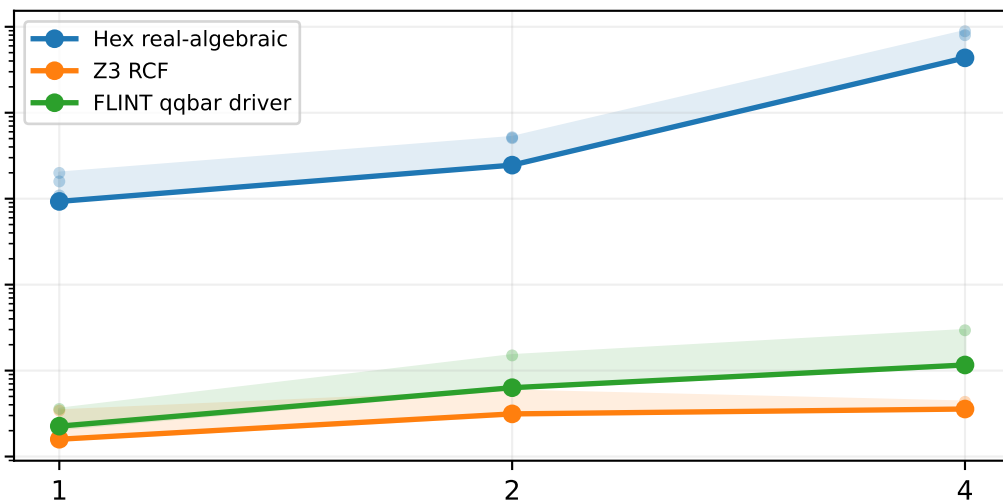
Hex / external (>1: Hex slower)



Polynomial degree n (coefficient height fixed)

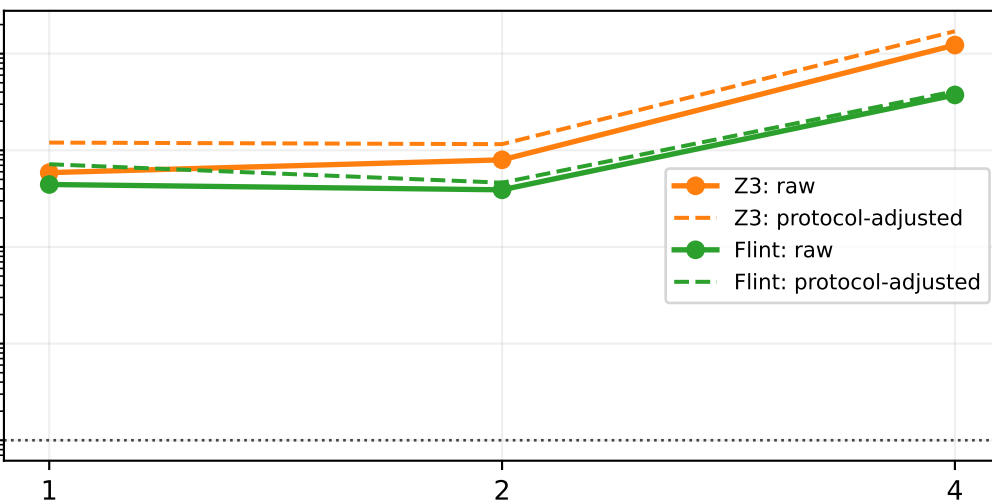
$$X^n - \sqrt{2} \text{ (n=1: one root; n} \geq 2 \text{: two roots)}$$

Root API + result checks (ms, log scale)



Polynomial degree n (coefficient height fixed)

Hex / external (>1: Hex slower)



Polynomial degree n (coefficient height fixed)

Shared host chungus2, leased CPU 38; 4 adjacent AB/BA pairs per backend/rung. All samples retained; no fitted model. Prepared coefficients; solving, ordering and fingerprints timed. External exact-annihilation checks, JSON and cleanup included. Pre-change unpinned degree-16 rational and degree-8 quadratic probes hit a 60 s whole-child cap; no numerical operation time inferred. Larger post-change whole-child probes pass in 9.3-9.4 s including preparation; they are not plotted as operation-only points. Source c6b821d7b; protocol adjustment is a framing control. Comparison failures/censored arms: 0; collected 144/144.